MIL-C-14817(MU)
21 October 1968

SUPERSEDING
FA(FC)-PD-71
9 May 1961

MILITARY SPECIFICATION

CONTROL, FREQUENCY: 10525510

1. SCOPE

1.1 This specification covers one type of frequency control used in the Radar Chronograph, M36. The assembly hereafter referred to as the AFC (Automatic Frequency Control), is used to generate the reflector voltage required to operate a klystron oscillator, and to control the klystron frequency by controlling the reflector voltage.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18, Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 1290

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MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army Frankford Arsenal

F10525510	Control, Frequency 10525510

PACKAGING DATA SHEET

10525510	Packaging of Control, Frequency

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication - The AFC shall be manufactured in accordance with Drawing D10525510 and drawings pertaining thereto.

3.2 General specifications - The AFC shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Workmanship

e. Soldering electrical connections

3.2.2 MIL-C-46357

a. Wiring

b. Threaded fasteners

c. Marking

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d. Finishes

e. Etched circuit assemblies

3.3 Environmental

3.3.1 Shock - The AFC shall be capable of continuous operation while being subjected to three (3) shocks of 50 gravity units (g's) of eleven (11) milli second duration in each of the following directions:

a. Bottom down

b. Front down

c. Right side down

d. Back down

e. Left side down

f. Top down

Subsequent to the above shock, the AFC shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.3.2 Vibration - The AFC shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz. The entire frequency range, from 10 to 55 hertz and return to 10 hertz, shall be uniformly traversed in 1 minute ± 15 seconds. This motion shall be applied for a period of thirty (30) minutes ± one (1) minute in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 1.5 hours. During the above vibration, the AFC output voltage (-800VDC) shall be continuous and uninterrupted. Subsequent to the above vibration, the AFC shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.3 Humidity - The AFC shall be capable of continuous operation during four (4) hours ± ten (10) minutes of continuous exposure to a relative humidity of ninety (90) to ninety-five (95) percent at a temperature of plus 100° ± 5°F. During and subsequent to the above humidity, the AFC shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.4 Sand and dust - The AFC shall be capable of withstanding three (3) hours of continuous exposure to blowing sand of particle size 0.01 to 1.00mm

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and blowing dust of particle size 0.0001 to 0.01mm with a density between 0.1 and 0.5 gram per cubic foot of dust at a temperature of plus 75° ± 5°F, a relative humidity between fifteen (15) to thirty (30) percent, and a wind velocity of fifteen (15) knots. Subsequent to the above sand and dust exposure, the AFC shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.5 Altitude

3.3.5.1 Operating - The AFC shall be capable of continuous operation during four (4) hours of continuous exposure to an altitude of 10,000 feet at a temperature of plus 75°F. During and subsequent to the above exposure, the AFC shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.5.2 Non-operating - The AFC shall be capable of withstanding four (4) hours of continuous exposure to an altitude of 50,000 feet at a temperature of minus 65°F. Subsequent to the above exposure, the AFC shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.6 Fungus - The AFC shall be capable of withstanding twenty-eight (28) days exposure to a composite spore suspension consisting of the following cultures:

a. Chaetomium globosum

b. Aspergillus flavus

c. Aspergillus versicalor

d. Penicillum luteum

e. Rhizopus nigricans

The composite spore suspension shall be incubated at an ambient temperature of 81°F ± 4°F with an RH of 87 ± 2%. Subsequent to the above exposure, the AFC shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.7 Storage temperatures - The AFC shall be capable of withstanding temperature stabilization at plus 160°F and minus 80°F for a period of eight

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(8) hours at each temperature extreme. Subsequent to each of the above temperature extremes (+160°F and -80°F) after temperature stabilization at +75 ± 15°F, the AFC shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.3.8 Operating temperatures - The AFC shall be capable of continuous operation for eight (8) hours at +140°F. The AFC shall also be capable of continuous operation for eight (8) hours at minus 50°F. During and subsequent to the above temperature extremes (+140°F and -50°F), the AFC shall not have been damaged or adversely affected and shall meet the requirements of 3.5 at least once at the beginning, middle, and ending of each 8 hour temperature extreme. There shall be a minimum of five (5) to six (6) minutes warm-up time prior to performing the first test of 3.5 at -50°F.

3.4 Electrical

3.4.1 Continuity - The AFC shall demonstrate continuity in accordance with the electrical schematic, Drawing F10525547

3.5 Performance

3.5.1 Input amplifier balance - The dc input amplifier shall be capable of being adjusted from minus 5 vdc through zero to plus 5 vdc. The balance potentiometer shall be adjusted to value between ±0.25 vdc. The maximum allowable drift shall not exceed plus or minus 3 vdc within the first 15 minutes of operation.

3.5.2 Output voltage - The output voltage shall be capable of being adjusted from minus 770 vdc to minus 830 vdc.

3.5.3 Control sensitivity - The output voltage shall deviate a minimum of plus and minus 10 vdc for an input voltage of minus and plus 5 mvdc, respectively.

3.5.4 Ripple - The ripple voltage on the output shall not exceed 30 mv peak-to-peak.

3.6 Continuity - There shall be continuity between P1-9 and P1-10.

3.7 Reliability run-in - In order to maximize component reliability, prior to acceptance by the Government each AFC shall be subject to the following run-in conditions:

a. Temperature run-in - The AFC without power applied, shall be temperature stabilized at plus 125°F and minus 25°F. At each temperature

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extreme, after temperature stabilization, power shall be applied and each unit shall be tested to all the performance requirements of 3.5 for a period of three (3) hours with a minimum of one (1) complete test to the requirements of paragraph 3.5 occurring at the beginning, middle and end of the three (3) hour period. Subsequent to the above tests, the AFC shall not have been damaged or adversely affected, and shall meet the requirements of paragraph 3.5.

b. Vibration run-in - The AFC shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of the(10) and fifty-five (55) hertz and return to 10 hertz in 1 minute. This motion shall be applied for a period of ten (10) minutes in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 0.5 hours. During the above vibration, the AFC output voltages (-800 VDC) shall be continuous and uninterrupted. Subsequent to the above vibration, the AFC shall not have been damaged or adversely affected, and shall meet the requirements of paragraph 3.5

c. Shock run-in - The AFC shall be capable of continuous operation while being subjected to three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

a. Bottom down

b. Front down

c. Right side down

d. Left side down

e. Back down

f. Top down

During the above shock, the AFC with power applied, shall comply with 3.5.2 (adj-voltage to -800vdc) and be monitored for a continuous uninterrupted output of -800vdc. Subsequent to the 18 shocks, the AFC shall be examined for damage, then tested as specified in 3.5

3.7.1 Failures during reliability run-in - Replacement of defective components and corrections of workmanship defects may be made during run-in under the following conditions:

a. Any part, module, component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the reliability run-in environmental requirements of the automatic frequency control.

b. Any replacement part shall have been exposed for at least one (1) hour at the extreme temperature being run immediately previous to being used as a replacement

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c. After a correction has been accomplished, at least one (1) hour of operation with a minimum of one (1) test to the requirements of 3.5 shall be successfully accomplished at the temperature at which the failure occurred.

d. A failure during 3.7b or 3.7c shall necessitate a complete rerun of the tests specified in 3.7.b and 3.7.c.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) AFC's manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection,

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4.3 Inspection provisions

4.3.1 Examination and tests

4.3.1.1 Components and subassemblies - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the applicable Quality Assurance Provisions of MIL-F-13926.

4.3.2 Acceptance and rejection - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling - At the beginning of production and subsequent to First Article Approval (see 4.2), three units shall be selected by a Government representative and two from each 50 produced. The sample shall have successfully met all other examinations and acceptance tests specified herein, and shall meet the following requirements and tests:

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
401. Fabrication	3.1	Applicable drawings and Specs - Visual
402. General Specifications	3.2	MIL-F-13962 and MIL-C-46357
403. Shock	3.3.1	4.6.1.1
404. Vibration	3.3.2	4.6.1.2
405. Humidity	3.3.3	4.6.1.3
406. Sand and dust	3.3.4	4.6.1.4
407. Altitude-operating	3.3.5.1	4.6.1.5
408. Altitude-non-operating	3.3.5.2	4.6.1.5.1
409. Storage temperatures	3.3.7	4.6.1.7.1
410. Operating temperatures	3.3.8	4.6.1.7.2

4.4.1 Failure of sample - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor, and resubmitted samples have been approved (see 4.3.2).

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4.5 Inspection equipment - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment

4.5.2.1 Contractor design - The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures

4.6.1 Environmental - Test equipment and test facilities utilized in the following environmental tests shall be in accordance with "Test Facilities" requirements of MIL-F-13926 and conditions specified in 3.3.

4.6.1.1 Shock - The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.3.1. During the above shock, the AFC with power applied, shall comply with 4.6.2.2 (adjust output voltage to -800vdc) and be monitored for a continuous uninterrupted output of -800vdc. Subsequent to the 18 shocks, the AFC shall be examined for damage, then tested as specified in 4.6.2.

4.6.1.2 Vibration - The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.3.2. Subsequent to the 1.5 hour vibration, the AFC shall be examined for damage, then tested as specified in 4.6.2.

4.6.1.3 Humidity - The humidity test shall be performed as specified in MIL-STD-810, Method 507, Procedure I for four (4) hours at the conditions

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specified in 3.3.3. Subsequent to this test, the AFC shall be examined for damage then tested as specified in 4.6.2.

4.6.1.4 Sand and dust - The sand and dust test shall be performed as specified in MIL-STD-810, Method 510 with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.3.4. Subsequent to this test, the AFC shall be examined for damage then tested as specified in 4.6.2.

4.6.1.5 Altitude-operating - The altitude test shall be performed as specified in MIL-F-13926, Procedure I at an altitude of 10,000 feet and as specified in 3.3.5.1. The AFC shall be exposed to the above condition for a time period required to allow all entrapped air in the chamber to escape as evidenced by chamber pressure stabilization. The barometric pressure shall then be maintained to simulate 10,000 feet (20.58 inches of mecury) altitude condition. While the AFC is maintained at this pressure for four (4) hours, the AFC shall be subjected to the tests specified in 4.6.2.

4.6.1.5.1 Altitude-non-operating - The AFC shall be subjected to barometric pressure specified in MIL-F-13926, Procedure II for four (4) hours as specified in 3.3.5.2. Subsequent to this test, the AFC shall be examined for damage, then tested as specified in 4.6.2.

4.6.1.6 Fungus - One (1) AFC shall be subjected to the fungus test specified in MIL-F-13926. At the end of the 28 day incubation period, the AFC shall be tested as specified in 4.6.2 for compliance with 3.3.6. Subsequent acceptance of AFC shall be by certification.

4.6.1.7 Temperature cycling

4.6.1.7.1 Storage temperatures - The AFC shall be placed in a chamber and the temperature gradually raised until thermally stabilized at +160°F for eight (8) hours and at -80°F for eight (8) hours. The AFC shall be returned to room ambient temperature +75 ± 15°F, examined for damage then tested as specified in 4.6.2.

4.6.1.7.2 Operating temperatures - The AFC shall be placed in a chamber and the temperature gradually raised until thermally stabilized at +140°F for eight (8) hours. The AFC shall be tested as specified in 4.6.2 during the first hour, at the four (4) hour period, and within the last hour of the eight (8) hour period. The temperature shall then be reduced to -50°F. When thermally stabilized at this temperature, the AFC shall be tested as specified in 4.6.2 at the three (3) time periods specified above. A five (5) to six (6) minute warm-up period shall be allowed prior to performance of the first test. Subsequent to above temperature extremes, the AFC shall not have been damaged or adversely affected and shall meet the requirements of 4.6.2.

4.6.2 Performance - Unless otherwise specified, the performance tests specified herein shall be performed on a 100% basis using the following input

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voltage combinations:

a. +22.5 VDC to PIN 8 of P1
-495 VDC to PIN 2 of P1
+183 to PIN 13 of P1
Ground to PIN 6 of P1

b. +21.5 VDC to PIN 8 of P1
-505 VDC to PIN 2 of P1
+175 VDC to PIN 13 of P1
Ground to PIN 6 of P1

4.6.2.1 Input Amplifier Balance - Insert a voltmeter between J1 and J2. Adjust BAL potentiometer (R10) to obtain a voltage from -5 VDC through zero to +5 VDC. Readjust BAL potentiometer to read 0 ± 0.25 VDC. The above voltage (0 ± 0.25 VDC) shall be monitored for fifteen (15) minutes in which the maximum allowable drift shall not exceed ±3 VDC.

4.6.2.2 Output voltages - The voltage output between J3 and ground shall be adjusted between -770 and -830 VDC when the AFC potentiometer (R17) is varied.

4.6.2.3 The above test shall be repeated monitoring P1-1 in place of J3.

4.6.2.4 Control sensitivity - Adjust AFC potentiometer R17 to obtain -800 ± 5 VDC at J3. With an input to P1-11 of minus 5 MVDC the above output at J3, shall have a minimum change of +10 VDC.

4.6.2.5 The above test shall be performed using an input to P1-11 of plus 5 MVDC and obtaining a minimum change at J3 of minus 10 VDC.

4.6.2.6 Ripple - The ripple voltage at J3 (-800 ± 10 VDC) shall not exceed 30 MV peak-to-peak, as measured with an oscilloscope.

4.6.3 Electrical

4.6.3.1 Continuity - Continuity shall be tested by standard measuring equipment for compliance with 3.5.

4.6.4 Reliability run-in - The following run-in shall be performed on a 100% basis.

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4.6.4.1 Temperature Run-in - The AFC shall be placed in a chamber and the temperature gradually raised until thermally stabilized at plus 125°F. After temperature stabilization, power shall be applied and maintained for three (3) hours. Five (5) minutes after power is applied, a complete test to the requirements of paragraph 4.6.2 shall commence. A second test to the requirements of paragraph 4.6.2 shall be accomplished during the second hour and a third test to the requirements of paragraph 4.6.2 shall be accomplished during the third hour of the three (3) hour test. Repeat the above tests at a temperature of minus 25°F. Subsequent to the above temperature extremes (plus 125°F and minus 25°F) the AFC shall not have been damaged or adversely affected and shall meet the requirements of paragraph 4.6.2.

4.6.4.2 Vibration Run-In - The vibration test shall be performed as specified in MIL-STD 810, Method 516 at the conditions specified in 3. 3.2 During the above vibration test, the AFC shall meet the performance test as specified in paragraph 4.6.2.2. Subsequent to the vibration test, the AFC shall be examined for damage and shall meet the performance requirements of paragraph 4.6.2.

4.6.4.3 Shock run-in - The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.3.1. During the above shock, the AFC with power applied, shall comply with 4.6.2.2 (adjust output voltage to -800VDC) and be monitored for a continuous uninterrupted output of -800VDC. Subsequent to the 18 shocks, the AFC shall be examined for damage, then tested as specified in 4.6.2.

4.6.4.4 Failures - Failures occurring during Reliability Run-in shall not be cause for rejection, but shall be corrected as outlined in paragraph 3.7.1.

4.6.5 Packaging, packing, and marking - Quality assurance provisions for packaging, packing, and marking shall be as specified in MIL-P-14232 and Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking - Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10512508. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended Use - The AFC is a subassembly used to generate the reflector voltage required to operate a klystron oscillator, and it controls the klystron frequency by controlling the reflector voltage.

6.2 Ordering Data - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging, and packing.

c. Applicable stock number.

d. Applicable packaging data sheet (see 5.1).

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Preparing Activity:

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Project Number 1290-A196

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